

CHAPTER 14

Attention and Impulsivity

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INTRODUCTION

Few abilities are as important for success in contemporary societies as the ability to exercise self-control. Conversely, few developmental difficulties generate as much controversy and concern in our society as problems with inattention and impulsivity.

Self-control and self-regulation (defined momentarily) develop, and go awry, via the synergistic interplay of multiple psychological, neural, behavioral, and social systems, including bottom-up reactive processes (i.e., behavioral inhibition and approach, in turn directly influencing arousal regulation), top-down effortful control or executive functions, language development, and dynamic

¹Color versions of Figures 14.1, 14.2, 14.4, 14.5, 14.6, 14.11 and 14.13 are available at <http://onlinelibrary.wiley.com/book/10.1002/9781119125556>

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interaction with socialization processes (e.g., parent-child relationships and learning opportunities) in infancy, preschool, and the school-age years. In the prior edition of this volume, my coauthors and I (Nigg, Hinshaw, & Huang-Pollack, 2006) emphasized the interplay of socialization and interior dynamics of the child's cognitive, language, and emotional development in the gradual unfolding of self-regulation, as well as in the breakdowns that emerge into attention-deficit/hyperactivity disorder (ADHD) (APA, 2013).

This chapter therefore takes a somewhat different tack. It does not recapitulate the review of socialization and the role of language. Instead, after noting basic framing concepts similar to those described by Nigg et al., including an updated account of clinical features and history, it elaborates a two-process conceptualization of self-control of attention and impulse. This chapter is therefore entirely

updated from the prior edition of this handbook. It outlines how this basic model can be used both to understand disorders like ADHD and to guide new research.

In the process, the chapter considers the formal constructs of attention and impulse control, outlining their development, and pointing out illustrative recent findings related to ADHD that help the reader understand normal and abnormal development of these abilities. The review is selective and heuristic, highlighting recent developments, selecting classic findings, and pointing readers to meta-analyses and comprehensive recent reviews on key component literatures for further study.

Developmental changes are notable throughout the life span, including infancy and young adulthood (Nigg et al., 2005) and old age (Hasher & Zacks, 1988; Zacks & Hasher, 1994). However, data on ADHD in later life are sparse; normative cognitive aging is beyond the scope of this chapter and well reviewed elsewhere (Park, 2012; Spreng, Wojtowicz, & Grady, 2010). This chapter therefore largely bypasses discussion of cognitive decline of attention or impulse control in later adulthood.

Throughout, in keeping with the developmental psychopathology perspective of multiple levels of analysis (Cicchetti & Dawson, 2002), the essay considers multiple perspectives. While emphasizing the psychological and neural levels of analysis, careful note is taken of considerations relevant to the genetic and socialization contexts, all with an eye on development.

Self-Regulation and ADHD

In the normative domain, one can invoke the concept of *self-control* (implying *deliberate* control or regulation of thoughts, behavior, and emotions) or the even broader domain of *self-regulation*, encompassing both deliberate self-control and involuntary regulatory processes related to attention, mental ideation, behavior, and emotion. Because in most research the two constructs are not operationally distinguished, the text herein refers to self-regulation except when the data clearly pertain only to self-control. The wide array of positive and negative life outcomes that are related to self-regulatory abilities includes physical health, mental health disorders, substance abuse, academic achievement, school readiness, learning, peer relations, development of conscience, resilience to adversity, and even mortality and longevity (Blair & Razza, 2007; Calkins & Keane, 2009; Clausen, 1993; Gorenstein & Newman, 1980; Iacono, Malone, & McGue, 2008; Kern, 2008; Krueger et al., 2002; Krueger, Markon, Patrick, & Iacono, 2005; Moffitt et al., 2011; Nigg, 2006a; Wood, Rijdsdijk, Asherson, & Kuntsi, 2009; Young et al., 2009). The importance of adaptive

and maladaptive self-regulation and its relevance to life outcome therefore is compelling.

In the maladaptive domain, self-regulation broadly defined is so fundamental that it influences almost all psychopathology to some extent, from depression, to aggression, to ADHD (Rudolph, Troop-Gordon, & Llewellyn, 2013). This includes poor regulation of attention, planning, foresight, and working memory (inattention) and poor regulation of decision making, motor response, speech, and emotional expression (impulsivity). Mirsky and Duncan (2001) listed 17 psychiatric and neurological conditions associated with disordered attention, including schizophrenia, autism, fetal alcohol syndrome, closed-head injury, and narcolepsy. Crucially, however, the term *attention* connotes multiple functions; not all such functions are equally affected across these various conditions.

Impulse control likewise is multiply defined and multicomponential (Nigg, 2000; Whiteside & Lynam, 2003) and appears in a range of conditions, of which ADHD may be the most prominent but is hardly the sole exemplar. Antisocial behaviors (including conduct problems and psychopathy), eating disorders, personality disorders (notably borderline personality disorder and antisocial personality disorder), substance abuse disorders, tic disorders, obsessive-compulsive disorder, bipolar disorder, and pathological gambling are only some of the psychopathologies readily associated with serious maldevelopment of impulse control (Coles, 1997). Many of these are syndromes primarily observed in adulthood, and their relation to childhood impulsivity is largely unstudied. Nonetheless, clearly the correspondence is not one-to-one between the topic of disorders of attention and impulse control and the clinical syndrome of ADHD.

Indeed, this fundamental quality of underlying shared dimensionality is part of the reason that there is extensive overlap between ADHD and aggression and some overlap between ADHD and mood problems such as irritability (Shaw, Stringaris, Nigg, & Leibenluft, 2014), as well as to a lesser extent mood disorders like depression. That is, the ability to regulate impulse and attention is related to fundamental human capabilities that cut across descriptive categories of psychopathology. To constrain the scope of discussion, however, and because aggression and mood problems are discussed in other chapters, this chapter's emphasis is ADHD, a syndrome which is paradigmatic of problems in the domain of self-regulation. Readers may bear in mind that the underlying regulatory abilities discussed herein are likely related across conditions other than ADHD.

At the same time, ADHD affords certain unique considerations. Most salient is the perspective taken in *DSM-5*

(APA, 2013), that ADHD is a neurodevelopmental condition. It is thought to have developmental roots in early in childhood and perhaps even in infancy (Auerbach et al., 2004; Sullivan et al., 2015). Thus, the maturational processes of self-regulation are of central importance. In early precursive form, preschooler problems with self-regulation and early school age problems with inattention and impulsivity (as well as hyperactivity), which typically co-occur with aggression, mood regulation, or depression (although, importantly, this is not the only route to depression). While the picture for girls is somewhat more complex, most boys with conduct disorder also previously had ADHD, and the combination of ADHD and conduct disorder is a primary route to drug addiction and alcoholism. Thus, shared neurobiological mechanisms are expected in ADHD, conduct disorder, and drug addiction. The Psychiatric Genetics Consortium pooled genome-wide data (from arrays that assess hundreds of thousands of common variants called single-nucleotide repeats or SNPs) with a sample of some 30,000 individuals from one of five psychiatric disorders, including ADHD. Risk genes were identified that tended to be common across disorders (Smoller et al., 2013), although there was some nonoverlap for ADHD with the other disorders. Among child disorders, many neuropsychological findings seen in ADHD are seen in other disorders also, though sometimes with variations that support partial distinctions among conditions (Seguin, Parent, Tremblay, & Zelazo, 2009).

As might be expected from these intersections, many proposals for the psychological and neurobiological

mechanisms involved in ADHD have been offered in the contemporary period (Barkley, 1997; Berger & Posner, 2000; Casey, Tottenham, & Fossella, 2002; Castellanos & Tannock, 2002; Douglas, 1972, 1999; Newman & Wallace, 1993; Nigg, 2001; Nigg, Swanson, & Hinshaw, 1997; Nigg, 2006; Sagvolden, Johansen, Aase, & Russell, 2005; Satterfield & Dawson, 1971; Sergeant, Oosterlaan, & van der Meere, 1999; Zentall & Zentall, 1983). Many are similar to what could be hypothesized for other disorders of impulse control. This nonspecificity is not belabored herein but should be borne in mind while considering genetic and neural findings in relation to inattention and hyperactivity-impulsivity or in relation to ADHD.

Overall, then, it is important to recognize that psychopathological categories exist in relation to underlying dimensions of liability, and also that they exist to some degree on a developmental continuum. Figure 14.1 shows one empirical estimate of how ADHD fits in with common types of psychopathology. It shows that hyperactivity-impulsivity tends to correlate strongly with oppositional defiance and conduct problems. Note, however, that even though inattention is placed on the externalizing factor in the figure, it is equally correlated with depression/anxiety symptoms. Also note that this helpful study by Lahey et al. (2008) did not consider the correlation of ADHD with learning disorders, autism, motor delays, or other neurodevelopmental conditions.

Few studies have done so. One study to consider this question, however, found that ADHD clustered strongly with ODD but also with autistic symptoms, motor coordination, and reading problems and that sibling cross

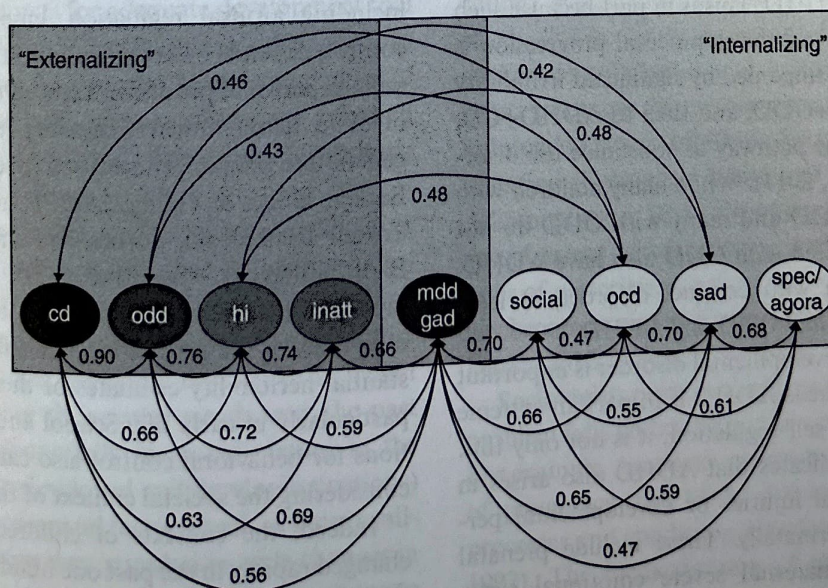


Figure 14.1 Correlations among child disorders. See footnote 1. Source: Reprinted with permission from Lahey, B. B., Rathouz, P. J., Van Hulle, C., Urbano, R. C., Krueger, R. F., Applegate, B., ... Waldman, I. D. (2008). Testing structural models of DSM-IV symptoms of common forms of child and adolescent psychopathology. *Journal of Abnormal Child Psychology*, 36, page 196.

correlations loaded better for autistic, motor, and reading problems than behavior problems (Couto et al., 2009). Indeed, when one considers the overlap among symptoms of these disorders, it is striking that ADHD symptoms have a substantial phenotype overlap with motor delays (Fliers et al., 2009; Piek, Pitcher, & Hay, 1999; Pitcher, Piek, & Hay, 2003; Rasmussen & Gillberg, 2000); phenotypic overlap with autism spectrum disorders, (ASD) in that 50% of children with ASD are hyperactive and about 20% of children with ADHD have sufficient social oddity to raise questions about ASD overlap (Matson, Rieske, & Williams, 2013; Reiersen, Constantino, & Todd, 2008); have shared heritability with autism (Lichtenstein, Carlstrom, Rastam, Gillberg, & Anckarsater, 2010; Musser et al., 2014; Rommelse, Franke, Geurts, Hartman, & Buitelaar, 2010); and have cognitive problems including lower average IQ (Nigg, Willcutt, Doyle, & Sonuga-Barke, 2005) and learning disabilities (Willcutt et al., 2010), with which there is again genetic overlap (Couto, et al., 2009; Loo et al., 2004; Willcutt et al., 2002). Additionally, ADHD is associated with early appearing alterations in neural development (Castellanos et al., 2002; Shaw et al., 2007). These early brain alterations are not fully explained by differences in the structure of the genome (Castellanos et al., 2003) and so are due at least in part to either stochastic epigenetic changes or environmental influences on gene expression. While interesting neural signatures have been linked to callous-unemotional (psychopathic) traits (Breedon, Cardinate, Lozier, VanMeter, & Marsh, 2015), it is relatively difficult to demonstrate similar magnitude developmental brain changes in children with ODD or CD who do not also have ADHD. This is in part because such children are rare. A typical developmental progression is from ADHD (often accompanied by significant irritability early in life) to ADHD+ODD, and then to ADHD+CD; from there emanates one pathway to substance use disorders (Molina & Pelham, 2014). While many children with ADHD do not have ODD and many with ODD do not develop CD, most children with ODD also have ADHD, and nearly all children with conduct disorder, at least among boys, previously had ODD and ADHD.

The idea of a neurodevelopmental disorder is important for another reason: whereas ADHD is in part an extreme of a dimension of poor self-regulation, it is not only this. Substantial evidence indicates that ADHD also arises in relation to subtle neural injuries or developmental perturbations pre- and perinatally. These include prenatal teratogenic exposure, maternal severe emotional stress during pregnancy, low birth weight, or premature birth, particularly when other developmental conditions such as

motor coordination problems or ASD symptoms, co-occur (Gillberg, 2003; Linnet et al., 2003; Nijmeijer et al., 2010; Owens & Hinshaw, 2013; Pearsall-Jones, Piek, Rigoli, Martin, & Levy, 2009; Sato, Aotani, Hattori, & Funato, 2004; Whitaker et al., 2011; Thapar et al., 2013). A caveat is that pre- and perinatal effects on outcome are difficult to detect after middle childhood (Wagner, Schmidt, Lemery-Chalfant, Leavitt, & Goldsmith, 2009) and are not necessarily causal. For example, maternal smoking in pregnancy seems to be largely a proxy for genetic risk, operating via gene-environment correlations (D'Onofrio et al., 2008; Obel et al., 2011; Thapar et al., 2009).

Even so, the weight of evidence indicates that in some instances ADHD is associated with poor self-control in the same way that a closed head injury is associated with poor self-control—that is, as a consequence of a subtle neural injury rather than as normal variation on a trait (Whitaker, et al., 2011). Thus, multiple etiological pathways must be borne in mind. In all, developmental paths to clinically significant inattention and impulse control problems are highly likely to involve both *equifinality*, meaning that several causal routes are contributory to ADHD, and *multifinality*, whereby early risks for ADHD may yield substantially divergent outcomes contingent on socialization and contextual processes (Cicchetti & Rogosch, 1996; Hinshaw, 1999).

Initial Comments on Etiological and Developmental Process

In the preschool years, the attainment of self-regulation, including arousal regulation, emotion, and attentional control, depends on socialization. It therefore is inherently an interpersonal process influenced by a complex interplay of child temperament (reactivity, presumably related to genetic and prenatal effects) and parent responsivity (Cecil, Barker, Jaffee, & Viding, 2012; Kim & Kochanska, 2012; Troxel, Trentacosta, Forbes, & Campbell, 2013; Ursache, Blair, Stifter, & Voegtline, 2013). In particular, family processes may be a primary mediator for the expression, over time, of the genetic mechanisms reflected in the substantial heritability estimates of these behavior problems, particularly in early life. School and community expectations for behavioral control also cannot be overlooked in considering the societal context of these problems.

Indeed, the contexts of children's development have changed rapidly in the past one hundred years in numerous ways beyond compulsory schooling. Diets have changed so much in the last century in the northern and western hemispheres that what most people in those societies

now eat has ceased to resemble the recent evolutionary diet, with complex effects on health that are still debated (Turner & Thompson, 2013; Walter & Ley, 2011). Hundreds of synthetic neurotoxins have flooded children's environments worldwide, including food, water, and air, with poorly understood consequences (Canfield, Gendle, & Cory-Slechta, 2004; Koger, Schettler, & Weiss, 2005; Lanphear, 2015; Nigg, 2006; Rice, 2000). In the last decade alone, the pace of technological change has been dramatic, continuing to change children's interpersonal and peer-relations throughout much of the world, through screen media exposure and social networking technology (Christakis et al., 2013; Sigman, 2012; Spies Shapiro & Margolin, 2013; Tandon, Zhou, Lozano, & Christakis, 2011). Political strife, upheaval, and violence have remained unfortunate realities for children in much of the world, often with substantial but poorly studied effects on the development of attention, self-control, and with them socialization, conscience, and interpersonal health (Betancourt et al., 2013; Betancourt et al., 2012). The meaning of ADHD in these varying contexts requires careful reflection, which this essay cannot fully address.

In the first world populations in which it has been primarily studied, ADHD (or its symptom domains) is highly heritable (Chang, Lichtenstein, Asherson, & Larsson, 2013; Kan et al., 2013). How do contextual and environmental pressures intersect with genetic effects? At times contextual forces are doubtless so powerful as to largely overpower genetic effects (e.g., an entire village of children forced to be child soldiers; Betancourt, et al., 2013). At other times, developmental contexts are sufficiently within the expectable limits for adequate development that genetic effects on individual variation are substantial. In short, when it comes to the complex behavioral picture of inattention and impulsivity, genetic effects are relevant, yet not directly causal or deterministic. Instead, two basic principles must be considered.

The first basic principal is that genetic and other child-specific mechanisms that influence individual differences in the expression of given traits or conditions (e.g., ADHD) do so within the context of effects on the entire population. For instance, individual differences in height are highly heritable, but the secular trends over the past century regarding substantially increased stature in the population at large (in developed and developing nations) are the result of environmental factors, such as more available calories. In the same way, population-wide changes in environment that influence level of a trait can contribute to children's attentional and regulatory problems regardless of heritability estimates within that population.

The second principle is that heritability estimates for nonmetabolic, multifactorial disease are better understood as heritability of liability than as heritability of disease. Put another way, genes likely influence susceptibility to psychiatric illness, while experiential events actualize that liability to varying degrees. This principle holds of liability even for infectious disease; heritability is substantial for tuberculosis, for instance (Fine, 1981) due to inheritance of vulnerability and protective qualities. Although specific considerations pertain to infectious disease, the same principle holds for behavior. This has been demonstrated in animal studies dramatically for some time. For example, Cairns and colleagues (Cairns, 1996) showed that in rodent populations bred to have nonoverlapping levels of stranger aggression, rearing in different socialization contexts fully eliminated this genetic effect in a single generation. This picture has helped drive strong interest in gene by environment (G×E) interactions, a theme later in this review.

Societal Context of Discussion: Medicalization, Controversies, Scientific Change

The clinical syndrome of ADHD and its treatment with medication has generated periodic and sustained controversy and social criticism for nearly half a century. This controversy requires comment to frame the current discussion. These societal discussions are provoked in part by ongoing secular increases in the numbers of children being identified with ADHD in the United States (Perou et al., 2013), although it should be noted that evidence of an actual increase in prevalence (as distinct from case identification), at least over the past two decades, is scant (Erskine et al., 2013). Western nations have also seen renewed concern about diversion of stimulant medication for non-prescription use for performance enhancement, leading to unintended health problems for users, which seizes media attention (Substance Abuse and Mental Health Services Administration, 2013; Tavernise, 2013). Clear data on the actual extent of such misuse, and whether it is related to rates of prescribing, is difficult to come by but nonetheless has become a significant concern in its own right (Kaye & Darke, 2012; Wilens et al., 2008; Weyandt et al., 2014).

Sociologists point out that *medicalization* occurs when a problem falls under the purview of the medical profession. For example, American society has seen the medicalization of pregnancy and birth, drinking problems, and other processes and conditions (Barsky & Borus, 1995; Conrad, 1992). The medicalization of attentional problems and hyperactivity, as well as disruptive behavior more generally, has long spurred critics to ask whether ADHD is a

valid psychiatric disorder (Searight & McLaren, 1998). It has also long sparked concern that rising medication rates may indicate an important shortage of other needed psychological or educational interventions for children (Safer & Krager, 1992) or that social control is being served rather than mental health (Menzies, 1997). The extreme forms of such criticism sometimes expressed are difficult to reconcile with the wealth of biological correlates of ADHD, from reliably reduced brain size (Castellanos, et al., 2002) from age 4 on, slower maturation of the cortical mantle (Shaw, Greenstein, et al., 2006), elevated levels of lead in their blood (Nigg, Nikolas, Kottnerus, Cavanagh, & Friderici, 2010), more family members with ADHD (Stawicki, Nigg, & von Eye, 2006), and differences in genotype (Banaschewski, Becker, Scherag, Franke, & Coghill, 2010).

The issue to which critics may be trying to allude is that biological markers are not sufficient for clinical diagnosis in ADHD or, indeed, in psychiatry generally. Diagnosis relies on clinical judgment about behavioral symptoms. That judgment is only as good as the skill of the clinician and the quality of the evidence base upon which she relies. The ontological status of ADHD as a real condition with a definable biological process or marker that can be measured in all or nearly all cases of the disease (like it presumably can in cancer or infectious disease) is still some distance off and may never occur. It is likely that if such diagnostic biomarkers are ever identified, they will not apply to all individuals currently diagnosed with ADHD.

Indeed, for all complex diseases (hypertension, obesity, diabetes, cancer, and psychiatric conditions) the ability to identify a single biological cause is likely to be possible only when we can differentiate subgroups (as in breast cancer recently, for example). Complex disease is inherently characterized by probabilistic prediction based on risk factors that predict morbidity and mortality. ADHD conveys considerable morbidity risk (Nigg, 2013a), and thus examination of probabilistic diagnostic categories remains useful and important.

Whether these biological arguments will take a new turn remains difficult to say. It is striking that the past decade has seen a sea change in available scientific tools: the field now has access to affordable and powerful genetic assays, new tools for integrated systems biology, more refined imaging measures that assay the entire brain and its organization instead of just regions of interest, internet based large scale data collection, and ever more powerful computers that all enable the era of big data to come to bear on the study of human development and behavior. Most of these tools are only beginning to influence our understanding of ADHD, but it can be anticipated that

they will do so to an increasing, if incremental, extent. With these considerations in mind, the discussion now turns directly to the syndrome of ADHD.

ATTENTION-DEFICIT/HYPERACTIVITY DISORDER: BACKGROUND

The changes in conceptual definition of and treatment rates for ADHD over the years are one contributor to controversy about its status and meaning. It therefore is helpful to briefly review key historical aspects. More detailed historical background is provided by Nigg, Hinshaw & Huang Pollock (2006), Taylor (2011) and Nigg and Barkley (Nigg & Barkley, 2014), from whom I borrow liberally to set a contemporary context.

History

Well before current conceptual models began to be developed, educators and clinicians began to notice clinical-level manifestations of inattention, impulsivity, and dysregulated behavior in school-age children. Description of disorders of attention first appeared in the medical literature a little over two hundred years ago. As noted by Barkley and Peters (2012), Weikart (in 1770 or 1775) described adults and children who were inattentive, distractible, impersistent, overactive and impulsive, reminiscent of today's description of ADHD. As detailed by Taylor (2011), Crichton (1798) described patients with extreme mental restlessness.

The first American record comes from the physician Benjamin Rush, often credited as the founder of American psychiatry, who in his pioneering textbook in 1812 mentioned a syndrome involving inability to focus attention (Rush, 1812). Other references occurred in Europe periodically during that century. Interest picked up considerably, however, in first part of the 20th century, perhaps in relation to the implementation of compulsory education in the 19th century. By 1908, the term minimal brain damage had entered the medical lexicon (Taylor, 2011) to represent a group of children displaying substantial ADHD-like behavioral disturbances yet without evidence of gross brain damage.

Following the worldwide influenza epidemic during and after World War I, some children developed encephalitis, and it was noted that many survivors of this infection developed significant problems in attentional deployment, regulation of impulses, and general behavioral control, leading to the term minimal brain damage syndrome, a

term later softened to minimal brain dysfunction, or MBD (e.g., Strauss, 1947). By the 1960s, the boundaries of MBD had become far too over inclusive (Clements & Peters, 1962; Wender, 1971), and the term began to fall into disfavor.

Others were calling for narrower and more descriptive appellations. Laufer and Denhoff (Laufer & Denhoff, 1957) described a hyperkinetic disorder of childhood, focusing more specifically on dysregulated motor behavior. The second edition of the *DSM* (APA, 1968) adopted this narrower conception and nomenclature.

An empirical tradition was also expanding, in which parent and teacher ratings of child problems were subjected to factor analyses in an attempt to discern identifiable, empirically supported syndromes of psychopathology (Quay, 1972; Achenbach, 1966; Achenbach & Edelbrock, 1978). Factors labeled hyperactivity and immaturity emerged in such dimensionalized accounts of child problem behavior. Indeed, prior to *DSM-III* (APA 1980), ADHD was identified by extreme scores on these newly emerging ratings scales such as the Conners Teacher rating scale (Achenbach & Edelbrock, 1978; Conners, 1969; Pelham, Fabiano, & Massetti, 2005).

Based largely on the research of Douglas and colleagues (Douglas, 1972), in the 1970s and 1980s researchers in North America began to conceptualize the core deficit of children considered hyperkinetic as related to dysregulated attention and impulse control rather than motor overactivity per se. By *DSM-III* (APA, 1980), in the United States the condition had been renamed attention deficit disorder, which was posited to consist of a subtype with hyperactivity and a second, albeit undefined, subtype without hyperactivity. Meanwhile, in Europe the World Health Organization *ICD-9* (World Health Organization, 1978) continued to refer to hyperkinesis and to use a more restrictive definition that excluded many children with comorbid conditions who were still included in the *DSM* definition.

The empirical tradition continued, with efforts to develop empirically based taxonomies of children that included dimensions related to hyperactivity and inattention (Achenbach & Edelbrock, 1983). By this time, stimulant medication treatments, which had originated decades earlier with the pioneering work of Bradley (1937), were generating controversy (Sroufe & Stewart, 1973), a theme that continued (Jacobvitz, Sroufe, Stewart, & Leffert, 1990; Safer & Krager, 1992) and persists, as noted earlier.

Renamed attention-deficit/hyperactivity disorder for *DSM-III-R*, *DSM-IV*, and *DSM-5* (American Psychiatric Association, 1987, 1994, 2013), the condition's core subtypes were expanded in *DSM-IV* to three: a Predominantly Inattentive type (similar to the *DSM-III* notion of attention

deficit disorder without hyperactivity), a predominantly hyperactive-impulsive type (unprecedented in previous nomenclatures), and a Combined type in which both symptom clusters were clinically salient. In *DSM-5* (APA, 2013), the structure of ADHD was largely unchanged although the subtypes were downgraded to presentations with text explaining their temporal instability.

Legislative changes in the early 1990s yielded the inclusion of ADHD as a potential trigger for special education services in public schools, while pharmaceutical companies began direct-to-consumer advertising, all increasing the salience of ADHD to families and clinicians. Prescription rates for stimulants, as for other psychotropic medications used with youth, doubled in the United States in the 1980s, repeated that in the 1990s (Safer & Krager, 1992; Safer & Zito, 1999; Zito et al., 2003), and continue to climb. Meanwhile, major clinical trials evidenced the efficacy of stimulant pharmacologic treatment for ADHD, if implemented carefully and with thorough monitoring, with combined medication-behavioral treatments yielding the highest likelihood of clinically meaningful improvement (Conners et al., 2001; MTA Cooperative Group, 1999). However, subsequent long term follow up studies indicate that the long term outcomes of brief intensive treatment were little altered (Molina et al., 2009), leading to an increased sense of the need to more fully understand pathophysiology and etiology.

Prospective longitudinal investigations were showing that ADHD was a condition likely to persist into adolescence in a majority of cases and to continue to yield impairment into adulthood in a substantial percentage (Mannuzza & Klein, 2000). Those findings led to the blossoming of research on ADHD in adults and the gradual recognition of the unique issues in establishing validity of the construct in adults, reflecting recognition of a research and clinical domain in ADHD that had been underway for a generation (Wood, Reimherr, Wender, & Johnson, 1976).

Thus, ADHD has for some time now been recognized as a condition that is not restricted to childhood or even adolescence. Yet the ongoing recognition of ADHD in adolescents and adults has in some ways exacerbated questions about the boundaries of normality when it comes to inattention and disorganization, as a harried, over-scheduled workforce and a student population driven by increasing competitive performance demands seeks relief and help.

In general, ADHD is currently understood as a clinically useful, statistically informative but not biologically meaningful cut point on a continuous dimension of behavior in the population (Larsson, Anckarsater, Rastam, Chang, & Lichtenstein, 2012; Levy, Hay, McStephen,

Wood, & Waldman, 1997; Levy, McStephen, & Hay, 2001; Willcutt, Pennington, & DeFries, 2000). It is provocative to consider whether this dimension we call ADHD in fact contains meaningful subconfigurations, which hew better to biological domains and that might isolate biologically more meaningful populations. That question is taken up in this chapter's concluding sections.

Clinical Features and Issues for ADHD

DSM-5 (APA, 2013) requires six of nine possible symptoms of inattention-disorganization or 6/9 symptoms of hyperactivity-impulsivity to be present for at least 6 months and to have onset prior to age 12 years. They must cause substantial interference in functioning and substantial features have to be present in more than one setting (e.g., home and school). The World Health Organization's *ICD-10* (World Health Organization, 1992) is expected to be updated with an 11th edition in 2017 or 2018, shortly after this volume is published, but likely will not have identical criteria to *DSM-5*, reflecting ongoing differences in how this syndrome is viewed in Europe and the United States. However, because most research on ADHD uses a *DSM* or else a dimensional perspective, the present chapter refers to ADHD without differentiating potential distinctions with hyperkinetic disorder in the *ICD*. The prevalence of ADHD worldwide is estimated by recent meta-analyses to be between 2.5% (Erskine, et al., 2013) using fairly conservative criteria, to about 5% using slightly broader criteria (Polanczyk, de Lima, Horta, Biederman, & Rohde, 2007; Willcutt, 2012), with a male preponderance on the order of 2.5:1 in community samples. A handful of key issues with the clinical construct are important to note.

The diagnosis of ADHD requires not only above-threshold numbers of symptoms but also (1) the early onset of these symptoms, (2) their developmental extremity (a key point, given that problems of inattention and impulsivity are ubiquitous in young children), (3) their cross-situational display (i.e., at home, at school, and often in the peer group or work setting), and (4) the requirement that they yield functional impairment. These guidelines have strong empirical support. Thus, assessment requires, at a minimum, the taking of a careful history; reports from multiple adult informants with well-normed rating instruments; differential diagnostic procedures to distinguish ADHD from either normal developmental variation or a host of medical and psychiatric conditions that feature inattention and impulse control problems in their presentation; and, when possible, direct observation of the child in family and school contexts. Considering functional

adjustment in multiple domains can assist with treatment tailoring (Pelham et al., 2005).

Willcutt et al. (2012) provided a thorough review of the empirical data regarding symptom dimensions and clinical subtypes, which yielded several important conclusions. First, a two-factor structure underlying the constituent symptomatology, corresponding to inattentive-disorganized and hyperactive-impulsive symptom domains, is valid and useful psychometrically. Both areas can be reliably measured by parent and teacher rating scale instruments. They show divergent validity, with inattentive behaviors uniquely associated with academic problems and hyperactive/impulsive behaviors aggregating with disruptive tendencies in school and home settings. (Factor analyses actually show that a three factor model, separating impulsivity from hyperactivity, fits slightly better but because the fit is excellent for two factors, this more parsimonious solution is preferred for most uses). Thus, the underlying dimensions of ADHD show considerable psychometric viability.

Second, however, the *DSM-IV* subtypes (nearly identical to the *DSM-5* presentation) had significant validity problems. For this reason, they were downgraded to presentations in *DSM-5*. These problems included lack of longitudinal stability (one type easily becomes another over time), lack of differential treatment response; and lack of differential clinical or cognitive correlates beyond those that are already attributable to the distinction between the inattentive and hyperactive symptom domains. However, almost no data have examined genetic or neuroimaging findings on putative subtypes or dimensions differentially. Family studies asking whether ADHD subtypes breed true have been mixed; one meta-analysis found a small but statistically significant familial separation of the combined and inattentive types (Stawicki, et al., 2006). Meantime, behavioral genetic analyses suggest that the HI type, unlike the others, is not heritable (Willcutt, et al., 2000); its validity in school-age children is doubtful (Willcutt, et al., 2012).

ADHD is heterogeneous and some form of subtyping is likely to eventually be useful, but this will probably entail revision if not overhaul of the existing presentations. In part, there remains considerable concern in the field about how to describe a group of children who appear inattentive and underactive, sometimes referred to as children with sluggish cognitive tempo (Barkley, 2013; Moruzzi, Rijdsdijk, & Battaglia, 2013). Many of these children are likely included in the ADHD inattentive presentation, but some experts suspect they have a distinct syndrome unlike other children in that presentation grouping. A better subtyping

scheme would reflect neurobiological variation better, and might also yield superior clinical stability and prediction.

The *DSM-5* criteria provide some parenthetical examples pertaining to adults, but probably are still inadequate to capture the syndrome in adulthood. The hyperactive-impulsive symptom cluster is likely to be particularly salient during the preschool and grade school years but fades in frequency and intensity by adolescence. On the other hand, the inattentive-disorganized symptom cluster, which becomes most recognizable with the advent of formal schooling, is more stable throughout adolescence (Hart, Lahey, Loeber, Applegate, & Frick, 1995) and may present as serious problems with self-organization and executive functioning in adulthood that are not fully captured in the current symptom list. Further, the range of impulsivity in adults is likely far more complex than captured in the three impulsivity items in *DSM-5*, as is noted in my discussion of impulsivity later. An established literature suggests empirical bases for refining the conception in adults (Achenbach, 1991; Conners, Erhardt, & Sparrow, 1999; Kessler et al., 2010), but this literature has not yet reached sufficient critical mass to overcome the advantages to the diagnostic nomenclature of retaining similar symptoms across age for ADHD.

New impairments unique to adolescents and adults must be considered including risky sex, drug experimentation, driving accidents, problems with employment, and marital conflict (Barkley, Fischer, Smallish, & Fletcher, 2004; Nigg, 2013a). A final complexity unique to the condition in adults concerns the potential for misdiagnosis in relation to Axis II psychopathology, or, in other words, the possibility that ADHD may develop into an enduring maladaptive personality style or trait by adulthood (Kernberg & Yeomans, 2013).

ETIOLOGIES

Genetics

Family studies established long ago that ADHD breeds true, with a twofold to fourfold increased risk among first-degree relatives see (Stawicki et al., 2006). How much of this familial similarity is due to genes versus common family experiences? Numerous twin and a few adoption studies from North America, Western Europe, and Oceania have established that at least in those cultures, and primarily relying on parent ratings, substantial portions of liability are carried by genetic variation, with a heritability coefficient, averaged across many studies, exceeding .8 (Boomsma, Cacioppo, Muthen, Asparouhov,

& Clark, 2007; Chang, et al., 2013; Grant et al., 2007; Kan, et al., 2013). Heritability estimates are lower, however, when teacher ratings and self-ratings are examined (Merwood et al., 2013), due to the fact that each twin has a different informant (Kan, van Beijsterveldt, Bartels, & Boomsma, 2014). Yet the heritability of a latent variable for shared parent and teacher agreement was .78 in one study (Derks, Hudziak, Van Beijsterveldt, Dolan, & Boomsma, 2006). Relatively few studies have examined twin concordance of ADHD diagnoses derived from full clinical evaluation or the combining of parent and teacher input on symptoms and impairment.

The variation in results for teacher vs. parent ratings raises questions of rater bias (known as contrast effects) as an influence on heritability estimates. Contrast bias (parents emphasizing differences more in DZ than MZ twins) inflates heritability estimates of activity level in preschoolers (Basque, Martel, Leduc, & Cantin, 2008). Such effects in ADHD ratings appear to depend on what rating scale is used. Reitveld et al. (2011) reported on a longitudinal study of a large sample of twins in Europe, with maternal CBCL ratings at four age points (3, 7, 10, and 12 years). Even with rater contrast effects controlled, heritability was above .7 at each age. Simonoff et al. (2011) confirmed maternal contrast effects but also noted biases in teacher ratings due to twin confusion (known as correlated errors), especially for MZ twins. In other words, many twins have the same teacher, and teachers have more difficulty keeping MZ twins straight in their minds. When these effects were accounted for, heritability was between .6 and .7 in developed nations. Inattention-disorganization and hyperactivity-impulsivity have partially distinct genetic influences (Nikolas & Burt, 2010), underscoring the importance of considering this two-domain structure in any statements about ADHD's etiology.

Nonshared environmental effects account for the remainder of variance in ADHD liability, which distinguishes it from conduct disorder (Burt, 2010). Thus, conduct disorder seems to have a greater component of shared environment than ADHD. This is consistent with a developmental model in which ADHD, a condition influenced by early genetic and nonshared environmental effects, gives rise via family and peer socialization processes to conduct disorder (Snyder, Schrepferman, Bullard, McEachern, & Patterson, 2012).

In response to these findings of high heritability for ADHD, researchers aggressively pursued molecular genetic studies during the period from the late 1990s to now. Some reflections on approaches to molecular genetic studies are in order to aid readers in evaluating his body of work.

In taking a molecular approach, one must consider the conceptual genetic model of the trait or disorder being studied. While this was largely guesswork in the case of psychopathology a decade ago, Figure 14.2 (Hindorff, Gillanders, & Manolio, 2011; Manolio et al., 2009) illustrates one general analysis that is influential in psychiatric and medical genetics of complex disease today. It shows that the magnitude of genetic effect on disease for individual genes grows smaller in a near logarithmic function as one proceeds from rare genes causing genetically simple diseases to common genes acting in concert to influence complex disease.

In the case of ADHD, the most common approach to studying molecular correlates in the past two decades was, from the viewpoint of this model, rather limited. The idea was to look for large effects in candidate genes using common gene variants. These were candidate genes because they were believed for theoretical reasons to be of interest, such as dopamine receptor genes. One early meta-analysis of that literature concluded that six genes have common markers that had been statistically associated with ADHD to date: dopamine transporter (DAT1), dopamine D4 and D5 receptors (DRD4, DRD5), serotonin transporter (5HTTLPR), HTR1B, and SNAP25 (Gizer et al., 2008). However, these in combination account for only 1% of phenotypic variance in ADHD and thus are not clinically useful, and their effects, in terms of our Figure 14.2, are small, with odds ratios around 1.1 (i.e., far too small to be of clinical interest, but potentially adding clues to pathophysiology overall, in conjunction with one another).

A second approach also assumes that ADHD is a product of common variants (the common disease + common variant model), but takes an agnostic approach to which genes are involved. The approach therefore scans the entire genome, called a genome-wide association (or linkage) study or GWAS. With this approach searches are conducted across hundreds of thousands of common markers (usually a particular type of marker called single nucleotide repeat or SNP) distributed across the genome. Somewhat to the surprise and disappointment of many scientists, genome-wide scans generally failed to identify important new genes in ADHD. However, when we consider Figure 14.2, in retrospect this result is not especially surprising. The model in Figure 14.2 predicts that the effects of individual variants are quite small. In GWAS, thousands of statistical tests are required, resulting in low statistical power to detect these small effects. Indeed, individual markers have not been seen reliably for most complex diseases or traits until sample sizes have reached into the range from 10,000 to 30,000 samples. Yet most studies of ADHD used only a few hundred to a few thousand samples.

Nonetheless, a few interesting findings emerged from GWAS of ADHD so far. Novel candidates may warrant follow up, including one that is under a genome-wide significant linkage peak in a meta-analysis, cadherin 13 (CDH13) (Lasky-Su et al., 2008). This gene is expressed in nicotinic receptors and involved in neurite outgrowth (Canino & Alegria, 2008). Further, specific genetic effects, particularly related to central nervous system development, are shared across several major psychiatric disorders (PGC,

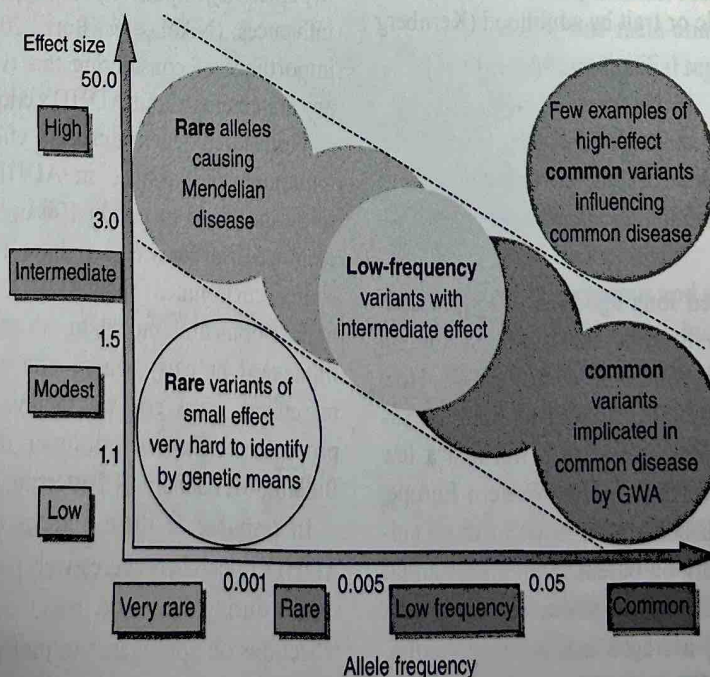


Figure 14.2 Molecular genetic meta-model. See footnote 1. Source: Reprinted with permission from T. A. Manolio, F. S. Collins, N. J. Cox, D. B. Goldstein, L. A. Hindorff, D. J. Hunter, ... P. M. Visscher (2009) Finding the missing heritability of complex diseases, *Nature*, 461(7265), page 749.

2013; Lotan et al., 2014) although ADHD does not fully share in this overlap (Smoller, et al., 2013).

Perhaps most interesting at the present juncture are analytic methods that manage to sum up the effects of all the variants on a chip, whether or not they are initially significant (for description of these methods, see Wray et al., 2014). This approach reveals that when all the common variants are taken into account, in aggregate they account for about 25% of the genetic variance in ADHD (PGC, 2013). Insights regarding molecular genetic overlap of ADHD with other traits and disorders are accordingly now possible and beginning to emerge at a rapid pace as of this writing (Hamshere et al., 2013; Martin, Hamshere, Stergiakouli, O'Donovan, & Thapar, 2015).

These polygenetic analyses confirm that common variants are important, but underscore that they do not act primarily via single gene effects but rather act in concert. That is, they likely operate in ADHD via their joint involvement in biological and metabolic systems. Those systems must now be described (next paragraph). Second, as corollary, other kinds of gene effects not due to common variants are carrying much, perhaps the majority, of the genetic variation in ADHD.

To build on the first of these hard-earned realizations, about the aggregate activity of common variants, will require organizing common gene variants into their chemical and physiological groupings, called gene-based pathways. This approach tests for significant association with an over- or underexpressed pathway and thus has more power than searches for individual markers. A small number of handful studies have attempted this approach with ADHD. Results, while preliminary, are intriguing, linking ADHD to gene pathways involved in synaptic transmission, nicotinic receptors (already one of the biochemical theories of ADHD), neuritic growth, and CNS development (Bralten et al., 2013; Cristino et al., 2014; Hammerschlag et al., 2014; Poelmans, Pauls, Buitelaar, & Franke, 2011; Stergiakouli et al., 2012; Yang et al., 2013). Whether these specific findings will reproduce remains to be seen, but it is likely that more gene-pathway-based approaches will be fruitful in the near future and will eventually converge on new insights.

To build on the related insight, about effects of noncommon variants, requires still another approach. Here, a different model is assumed. As also illustrated in Figure 14.2, instead of a common disease + common variant model, one now adopts a common disease + rare variant model. Here the assumption is that numerous distinct, but individually rare, genetic events are all giving rise to the same syndrome. The goal is to discover some of these rare structural variants, in hopes of gleaning new insights into pathophysiology that may apply to the syndrome more broadly.

Many of these are rare copy number variants (meaning the only difference in the allele is that a given nucleotide sequence is repeated too many times). They can be found by re-analyzing GWAS data. These studies are beginning to emerge on a regular basis and implicate genes involved in neural development, suggesting the beginnings of genetic clues to pathophysiology. Studies of this type are emerging at a steady pace (Elia et al., 2010, 2011; Jacob et al., 2013; Jarick et al., 2014; Lionel et al., 2011; Martin, Hamshere, Stergiakouli, O'Donovan, & Thapar, 2014; Williams et al., 2010). It has been hypothesized that 10–15% of the genetic variation in ADHD may be due to these CNVs. To date, work has been limited because only fairly large variants could be seen to date. As the ability to see smaller perturbations in the genome emerges, more findings in this area will emerge.

Still other rare variants can be discovered by looking within family pedigrees and sequencing exomal regions of the genome (nonsynonymous variants) or by sequencing the entire genome. Both methodologies are on the cusp of widespread application. Novel rare variants are expected, and likely to be discovered, in the process, but each will pertain only to a small fraction of the population with ADHD. If enough are discovered that have large effects, clinicians can test for them, but the real payoff may be in determining their pathophysiological activity to discover more about how ADHD develops in the brain more generally.

Last but far from least, the unexplained genetic variation in attention deficits, impulsivity, and dysregulation may be due to gene $\times$ environment interaction or epigenetic effects. This idea is of central importance and the implications are substantial from a multilevel, developmental psychopathology perspective.

Gene–Environment Effects

In the past decade, studies of gene–environment (G $\times$ E) effects have become standard in research on development, developmental psychopathology, and clinical disorders. Ever more sophisticated process models are now available, and G $\times$ E is approaching the level of a new paradigm for behavioral studies generally, with entire special issues of major clinical and scientific psychopathology journals devoted to G $\times$ E in the last couple of years (Nigg, 2013a, 2013b).

The still emerging literature on G $\times$ E effects on inattention, impulsivity, activity, or ADHD is promising. An early review (Nigg, Nikolas, & Burt, 2010) indicated reliable and consistent interactions of psychosocial distress measures and genotype for particular candidate genes, notably serotonin transporter and dopamine transporter. Subsequent

work continues to generally support this picture using a range of measures of psychosocial stress and expanding to other candidate genes (Berry, Deater-Deckard, McCartney, Wang, & Petrill, 2013; Elmore, Nigg, Friderici, Jernigan, & Nikolas, 2015; Li & Lee, 2012, 2013; Riva, Marino, Giorda, Molteni, & Nobile, 2015; Sanchez-Mora et al., 2015; van der Meer et al., 2014; van der Meer et al., 2015). However, these studies have generally relied on one or two candidate genetic markers in relation to selected measures of the environment. The hazards of such studies are many. In particular, (1) the environmental measures may be influenced by unmeasured genes and (2) if the allele frequencies are sub-optimal or environmental measures not properly scaled, statistical artifact can easily result (Eaves et al., 2000). Nonetheless, the general picture is that G×E effects are gaining considerable support as a key mechanism in ADHD etiology.

While better modeling will be needed to identify particular genes and particular environments and how they interact, the general approach already shows that larger genetic effects can be seen when looking within different environments. Now, with the existence of G×E established in specific instances and widely accepted conceptually, a particularly salient question for the G×E effect in the development of ADHD as well as normative self-control and self-regulation concerns the nature of the interplay. The idea that different forms of interaction exist and can be distinguished was first proposed over two decades ago by Aiken and West (1991), and has now been rediscovered in the area of G×E for a range of child characteristics (Belsky, Pluess, & Widaman, 2013; Pluess & Belsky, 2011). Figure 14.3, adapted from a recent paper in this genre, illustrates the central question.

The top panel of Figure 14.3 schematizes a vulnerability model. In some environments, genetic risk is not activated, but in other environments it is. This model, consistent with a classic diathesis stress model, has long been the best, if underutilized, general model in psychopathology. The bottom panel illustrates an alternative, which is called a susceptibility or plasticity effect. Here, the clinical genotype is actually a genotype that is maximally responsive to the environment. The genotype confers not vulnerability but *responsivity*. Under negative environments, one sees the typical poor outcome for the risk genotype. But in advantageous environments, the same genotype can have a superior outcome to the alternative genotype.

Numerous hazards accrue to attempting to statistically distinguish these effects. To name just one obvious one, the top panel, extended to infinity, would presumably

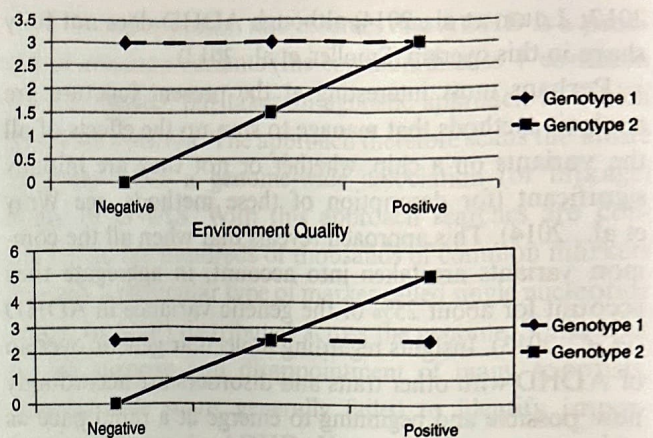


Figure 14.3 Vulnerability versus susceptibility models of G×E effect. Source: Adapted with permission from J. Belsky, M. Pluess, & K. F. Widaman, (2013) Confirmatory and competitive evaluation of alternative gene–environment interaction hypotheses, *Journal of Child Psychology and Psychiatry*, 25, page 1136.

eventually show the cross over pattern. However, at a practical level, this and other problems may be addressed with appropriate scaling, consideration of range, and other methods. Now, empirical studies have begun to distinguish plasticity from vulnerability effects in the area of parenting, temperament, and outcomes related in self-regulation of attention and impulse in early childhood (Kochanska & Kim, 2013b; Kochanska, Kim, Barry, & Philibert, 2011).

Further, recent years have seen exciting developments in epigenetics, that is, the recognition that experience can alter the genome—and thus the phenotype—sometimes dramatically. This represents at least one mechanism by which G×E can occur. Epigenetic modification occurs in several ways; one well-known way is through DNA methylation, which entails modification of chromatin, the material in which DNA is housed, to alter gene expression. That is, the expression of much of human variation may not depend only on DNA structure but on the regulatory markings that control whether and how a gene is expressed. Because these mechanisms remain speculative for ADHD, attention, and impulsivity, I do not elaborate further here. However, epigenetic mechanisms may prove extremely important in understanding human behavioral and cognitive development so I return to them later when considering future directions.

These two insights (the importance of G×E and the importance of epigenetic effects) have sparked a renaissance in studies of environmental contributions to ADHD (as well as several other psychiatric conditions) that is changing the face of research in the second decade of

the twenty-first century and may potentially also change the face of clinical practice in the decades that follow. Therefore, the role of the environment in the development of self-regulation, self-control, and ADHD is likely to be much more important than was widely believed ten or twenty years ago. However, this will be a modified understanding from even earlier days when ADHD was assumed to be entirely environmental. Instead of looking at environmental main effects, the field will undertake more specification of for whom (genetically) environments are exerting their effects on development and how (by modifying gene expression) they are doing it.

When it comes to the environmental effects, they must be considered at multiple levels of analysis and stages of development. We can consider the prenatal environment, where maternal diet and maternal stress likely exert influences on the developing fetus. We also can consider the early socialization environment. One interesting gap in our knowledge concerns very early socialization influences on later ADHD. The belief has been that disrupted parenting and discordant parent-child interactions result from, rather than cause, temperamentally mediated child noncompliance and impulsivity. This assumption is well supported by existing data, although problematic parent-child relationships may well be significant maintaining and exacerbating factors related to ADHD, yielding reciprocal, transactional chains of influence in conjunction with the child's core symptomatology (Hinshaw, 1999; Johnston & Mash, 2001). However, nearly all of the studies that support that view were conducted with children beyond the toddler years. Essentially nothing is known about the role of parent-child interchanges in the first year or two of life in ADHD.

ATTENTION AND IMPULSE CONTROL IN DEVELOPMENT: DUAL-PROCESS PERSPECTIVE

In thinking about how attention and impulse control develop, a heuristic model is helpful to organize the vast array of ideas and information available. Here one can gain considerable insight from a fundamental and ubiquitous approach in psychology, known as a hierarchical two-process model. Later it will be noted that more than two processes are involved ADHD. Nonetheless, while this is not a formal model, it provides substantial help with conceptualizing and organizing work in this field.

TABLE 14.1 Type 1 and Type 2 Regulation and Commonly Used Terminology

Type 1	Type 2
Reactive	Effortful
Automatic	Deliberate
Fast	Slow
Incentive motivated	Goal motivated
Hot	Cool
Bottom-up	Top-down
Develop early	Develop late

Two-process models must include the following elements: (1) type 1 process, (2) type 2 process, and (3) resource capacity or motivation² that influence the decision set points for which process will engage—in this case, the set point at which Process 2 is invoked to override Process 1. Models of this general nature, with varying specifics, are found historically in the ADHD literature (Nigg, 2001; Nigg, 2000b; Sergeant, et al., 1999; Sonuga-Barke, 2005; Sonuga-Barke, Dalen, Daley, & Remington, 2002), in theories of personality and temperament that relate to self-regulation (Block & Block, 1980) and are in concert with cognitive and social psychology models dating back over 30 years, as several excellent reviews attest (Evans, 2008; Evans & Frankish, 2009; Evans & Stanovich, 2013; Kahneman, 2011).

The perspective herein builds on earlier work by the author and collaborators (Nigg & Casey, 2005; Nigg, 2001, 2000b, 2006), complemented by related work by others (e.g., Sonuga-Barke, 1998; 2002), introduces some modifications in light of recent work in cognition (Evans & Stanovich, 2013), and uses this to describe a heuristic model that can usefully guide thinking both on ADHD and on the overall idea of self-regulation of attention and impulse and its development. From this perspective, the developmental psychopathology story is organized as it relates to disturbances in self-regulation and to ADHD.

The model has two distinct processes; they can be thought of as two aspects of a single system or as two systems (while experts in fact debate this last nuance, for present purposes either view is acceptable). These are termed type 1 and type 2 processes and heuristically summarized by their common appellations in Table 14.1.

Type 1 processes provide automatic handling of routine information, emergency reaction, and alert us to salient and unexpected information. Type 1 processing is the pre-

²The capacity or motivation aspect is also part of the model if it is recast as a single-process models with two aspects (Kruglanski, 2012).

sumed default: most behavior is automatic, presumably to conserve the organism's energy. Type 2 processes override stimulus-triggered responses and allow us to regulate our focus and our action in response. How does the mind determine whether to follow type 1 processing or to invoke the more costly and resource-intensive type 2 processing? This determination must take into account both the incentive value of the problem (motivation) and overall available resources (energy, arousal, and intellectual or working memory capacity). Therefore, the model must also include capacity or resource constraints that influence relative involvement of type 2 processes in a given situation.

Proposals for what goes wrong in ADHD invariably invoke a breakdown in type 1, type 2, or resource-capacity domains (or combinations thereof) to explain the syndrome. A critical question for theory of ADHD is, therefore, this: does regulatory failure occur because the type 2 process, once invoked, was ineffective? If so, was this ineffectiveness due to insufficient capacity or resources or to some other problem? Or does failure occur because the type 1 process failed to function properly and did not invoke the type 2 process in the first place? Theories of ADHD can be grouped according to which of these scenarios they emphasize.

Type 1 Process

Type 1 regulation is characterized by relatively automatic or autonomous processing. It is stimulus driven, rapid, occurs without intent or volition, and requires minimal working memory resources or application of attentional control. As conceived in relation to attention and impulse control, it is implemented primarily in posterior and subcortical neural networks, although functionally connected frontal brain areas can also be involved. It can interrupt deliberate activity, or be suppressed by it, but it is distinguishable from deliberate mental activity. An example of type 1 regulation is when behavior is automatically interrupted in response to a surprising sound or a frightening event (an example of subcortical activation sending involuntary signals into the attentional sphere and interrupting working memory and motor programs; (Gray, 1982; Gray & McNaughton, 1996; Kagan, Reznick, & Snidman, 1987), or when attention is automatically captured by a sudden movement (an example of a ventral attention network sending signals to the cognitive control system). In each case, attention automatically reorients to the new event, and other behavioral routines are interrupted. Thus, when she notices something anxiety provoking, a child may direct her attention to the concerning event or

person—reallocating and narrowing the focus of attention (Derryberry & Tucker, 1994).

An important principal in a dual-process account of self-regulation or ADHD entails the intersection of emotion and cognition (Nigg & Casey, 2005). In many psychology writings, type 1 processing is associated with relatively more affective response and type 2 with relatively less affective response and more cognitive processing or reasoning. While that summary has some value, readers are encouraged to think integratively about how emotion and cognition interplay in the moment-to-moment regulation of behavior. That is, emotion-free cognition and cognition-free emotion do not really occur. Rather, we are referring to the relative strength of emotional arousal.

Put another way, the incentives that shape attention can be sufficiently immediate as to triggering "hot" type 1 processing, or sufficiently distal as to require "cool" or type 2 strategic processing to attain them. Thus, even though attention is usually viewed as a cognitive process, it is just as often (if not always, to some degree) motivated by incentive concerns. These same incentives can also influence impulse control, leading either to a very strong impulse to action (e.g., excitement from an imminent reward) or to inhibiting some or all behavior (e.g., confrontation with an anxiety-provoking or unfamiliar situation). Such motivated responses are often relatively spontaneous, and they are relatively more automatic than the more deliberate responding characteristic of effortful control. In short, the type 1 response, often called the prepotent response in ADHD theories, can arise for multiple reasons—emotional salience (fear or excitement, which are associated heuristically with limbic circuits, in particular the amygdala), overlearning (associative or implicit learning, which can be associated in part with striatal structures), or response set (multiple prior instances of an overpracticed response). In each case, the response is activated without intent. It has become, or always been, automatized.

An interesting possibility, though largely bypassed here, is whether children with ADHD have difficulty automatizing behavior (Huang-Pollock & Karalunas, 2010). That is, in ADHD self-regulation may remain excessively dependent on type 2 process when it should not, hurting overall regulatory efficiency and causing effortful tasks to be overwhelming. An analogy would be poor readers for whom each word is effortfully sounded out rather than effortlessly recognized.

Type 2 Process

Type 2 regulation is characterized by demand on working memory or attentional resources, is experienced

subjectively as effortful or deliberate, is controlled, and is capacity limited. Its outcome is not automatic, but flexible (see thoughtful discussion by (Thompson, 2013). An example of such regulation is when I consciously bite my tongue to avoid interrupting a long-winded colleague so as to keep a New Years' resolution to be more patient. It is implemented to an important extent in prefrontal cortex, although it also involves the anterior cingulate cortex, and several other cortical and subcortical brain areas become involved due to their functional connection to prefrontal cortex. It serves to prime, bias, direct, or suppress the type 1 processes. This suppression is hypothesized to be implemented neurally by exertion of inhibitory projections backward to parietal cortex and downward, particularly via frontal-limbic-thalamic and frontal-striatal-thalamic and cortical-subcortical neural circuits.

Type 2 processing can be linked with productive concepts from the study of preschool temperament that refer to *effortful control*—the ability to delay or interrupt task-inappropriate behavior (Rothbart & Bates, 1998), or to assign positive motivation to a required task (Kochanska, et al., 2011). Effortful control shows longitudinal stability, with high scores for this ability associated with development of adaptive behaviors (Kochanska, Murray, & Coy, 1997; Kochanska, Murray, & Harlan, 2000), it also was helpful associations to developmental psychopathology, which help us integrate multiple literatures. Thus, excessively *high* control is associated with internalizing problems, while *low* control is associated with externalizing behavior problems over the preschool years (Murray & Kochanska, 2002).

Type 2 processing often requires working memory, and so is related with the idea of executive functioning, which has been a common neuropsychological approach to conceptualizing ADHD. It is widely agreed both that executive functioning is central to understanding attention and impulse control and that it is an underspecified construct (Pennington & Ozonoff, 1996; Zelazo et al., 2003). In a neuropsychological framework, executive function refers to strategic or relatively deliberate regulation of response to context and maintenance of behavior in relation to goals (Pennington & Ozonoff, 1996). While often supplanted by the more contemporary term cognitive control, the concept of executive function has additional connotations. It seems to reflect several component operations of which working memory is one; other important abilities include the capacity to temporally sequence information (Castellanos & Tannock, 2002; Fuster, 1997)—part of a broad but not a narrow conception of working memory (Baddeley, 2012), the ability to suppress prepotent responses whether

or not information is being held in working memory, and the ability to shift set. Numerous proposals for how to componentialize executive functions testify to its historical under-specification as well as its substantial heuristic utility (Barkley, 1997; Dempster, 1993; Engle, 2002; Friedman & Miyake, 2004; Logan, Cowan, & Davis, 1984; Lyon & Krasnegor, 1996; Mirsky & Duncan, 2001; Miyake et al., 2000; Monsell & Driver, 1998; Nigg, 2000b; Pennington & Ozonoff, 1996). This reflection on executive function also underscores the close relationship of the neural bases of deliberate control of attention and impulse (as well as the regulation of emotion) (Nigg & Casey, 2005; Posner & Rothbart, 2000).

With regard to impulse control, similar frontal-subcortical neural circuits are implicated in suppressing responses (Casey, et al., 2002). For example, suppressing inappropriate motor response during complex behavior appears to involve the dorsolateral prefrontal cortex, at least in young children (Diamond, Prevot, Callender, & Druin, 1997), as well as the orbitofrontal cortex (Aron, Fletcher, Bullmore, Sahakian, & Robbins, 2003; Posner & DiGirolamo, 1998) and the subcortical connections to each. Indeed, the subcortical and striatal structures to which these prefrontal regions are connected may be crucial in the actual suppression of a response (Kimberg & Farah, 1998) at least in certain contexts (Nigg & Casey, 2005). A developmentally oriented process conception of executive functioning emphasizes deliberate problem-solving steps, such as representing and manipulating the problem in working memory (planning), carrying out an intended action rather than a competing action, and detecting mismatch to expected results or success (Zelazo et al., 2003). Those functions likewise have clear relevance to attention and impulse control and all refer to type 2 regulatory processing.

Development, Socialization, and the Two-Process Model

The development of these neural and behavioral systems is in constant interplay with the child's interpersonal world as elaborated in more detail by us in the second edition of this handbook (Nigg, Hinshaw, & Huang-Pollock, 2006) via mechanisms like parenting attunement, verbalization of affect, and subsequent routinization, limit setting, and other activities, all of which operate in complex interchange with child reactivity and control to shape consolidation of self-regulation (Choe, Olson, & Sameroff, 2013; Kochanska & Kim, 2013a, 2013b; Koenig, Barry, & Kochanska, 2010). Neural development and self-regulation cannot develop apart from the early

parent-child relationship. Indeed for the first few months of life the mother provides the type 2 processing for the child and is the child's regulatory system in many respects. Scaffolding by caregivers in the early rearing environment, the modulation of stress, and the success of early learning all emerge as important in the early developmental consolidation of both type 1 but particularly type 2 self-regulation (Barry, Kochanska, & Philibert, 2008; Cecil et al., 2012; Kochanska & Kim, 2013b; Koenig, et al., 2010). Furthermore, it is likely that the aforementioned gene $\times$ environment interactions affect how parenting and child temperament interplay (Rothbart & Posner, 2005; Sheese, Rothbart, Voelker, & Posner, 2012). In short, while the current essay elaborates the within-child components of cognition and self-control, readers should be cognizant that interactions and transactions between within-child factors and environmental influences are crucial in any complete account of how self-regulation, inattention, or impulsivity develop. The transactional account is constrained in this chapter due to space, not due to its developmental importance.

Type 1 processes mature more rapidly than type 2 processes, commensurate with the brain's developmental maturation from rostral to caudal and the importance of prefrontal cortex in the implementation of advanced type 2 processing. Thus, behavior of young children is relatively less governed by type 2 processing than behavior of adults. Behavior of toddlers reflects relatively more precursive type 2 or effortful self-regulation than behavior of infants.

Infants allocate their attention (and their motor responses) simply on the basis of stimulus response and reflex response, which are largely midbrain-mediated processes that correspond to the reflexive attentional mechanisms discussed earlier. However, by later in infancy, sometime in the 1st year of life, children begin to respond to cues for incentives by directing their attention and making motor responses such as approaching a potential reward or withdrawing from a potentially distressing signal (Rothbart & Rueda, 2005). These latter responses correspond to affectively based type 1 allocation of attention and control of behavior. By the beginning of the toddler period, roughly about the age of 14 months, infants can be observed to deliberately turn their attention away from upsetting material, exemplifying a precursor to deliberate regulation of behavior and affect by executive control, and thus the beginnings of self-control (Rothbart & Bates, 1998).

In the toddler years, type 2 control of attention and impulse, in the guise of effortful control, develops rapidly. By about 30 months of age, children can begin to inhibit stimulus-driven motor response (Diamond et al., 1997;

Rothbart & Posner, 2005). For example, by this age, they can engage in interference control, go/no go, and stopping tasks (Diamond et al., 1997). Diamond and colleagues documented a steady development in the ability to inhibit competing, stimulus-driven motor responses by ages 3, 4, and 5 years; similar work is reported by Zelazo and colleagues (2003).

In middle childhood, many aspects of type 1 regulation mature, while type 2 regulation continues to develop. Huang-Pollock, Carr, and Nigg (2002) compared children in second through sixth grades on a computerized selection task evaluating two kinds of attentional selection. Late selection was progressively more mature in the older groups. Even sixth-graders were still well below adult level capabilities, although they were notably better than fourth graders. In contrast, early perceptual selection was essentially fully mature, at adult levels, even in the second-graders (Huang-Pollock et al., 2002). Similarly, Reuda et al. (2004) examined exogenous orienting (type 1), alerting (arousal) and strategic orienting (type 2). The exogenous orienting was essentially fully developed by the 6-9-year-old age period. All of these data are consistent with the proposition that relatively automatic type 1 processes develop earlier and complete their development at a much younger age than do cognitively demanding type 2 processes.

This general pattern of development mirrors the nonlinear and dynamic development of the brain systems involved. In particular, the cortical mantle develops in a partial caudal-rostral gradient, but with a peripheral-to-central tendency, so that the most anterior and posterior portions of the cortex mature first, followed by regions of prefrontal cortex. In addition, myelination as well as pruning of neural circuits to improve their specialization and efficiency proceed all the way into early adulthood (Benes, 2001).

In adolescence, the picture becomes more complex because cortical and subcortical brain regions and their supported capabilities are maturing in nonlinear, dynamic fashion. Thus, important changes in type 2 processes and regulation occur during adolescence. Here some aspects of controlled attention reach adult levels, but many aspects of impulse control do not. In particular, it is interesting that cool executive function or response control without strong emotional or incentive content improves in linear fashion and reaches adult levels by about 20 years of age (Hare et al., 2008). However, when hot decision making is involved (imminent emotional information and in particular, social cues of emotional salience), then adolescents have more failures than children or adults

(Casey & Caudle, 2013; Casey et al., 2011). Thus, adolescents have a unique impulsivity around socially heightened reward or reward-salient cues that will alter the risk profile for those with ADHD or with propensity to impulse control disorder. At the same time, in adolescence demands for cognitive control increase (e.g., multiple class schedules in high school) that require more executive functioning; those who are weak in this area may start to see more clinical impairment. Hyperactivity normatively levels off in adolescence (an ADHD adolescent does not run around the room, the way a 6 year old with ADHD does).

In turn, as the child matures in the social context, the strength of this two-process self-regulation system is likely to influence the child's capacities to be normally socialized by average expectable parenting and peer environments (Cicchetti & Lynch, 1995) and achieve adequate socialization adaptively in adverse or suboptimal caregiving environments. The import here is that if a child has very strong reactive processes (e.g., extremely strong negative affect), these may interfere with development of effortful control, in turn further weakening ability to regulate the reactive affects. Conversely, weak effortful control as an early temperamental characteristic could lead to difficulty with affect regulation as part of a two-way mutual influence process. This formulation may help to explain why executive and affective difficulties so often co-occur in children with disruptive behavior problems and ADHD.

Additional Considerations

Hierarchy Model of Complexity

This last point highlights that type 2 processes are not all unitary, but rather exist on a hierarchy of complexity. Representing a stimulus-response association is less abstract than remembering that the association only applies in certain contexts, that is, a meta rule. Likewise, considering an action's consequences in the next few minutes or hours is not the same operation, nor does it rely on the same neural bases, as considering consequences over much longer time periods. It can be argued that in the brain a rostral-caudal control gradient organizes type 2 or control processes, with the most anterior aspects of PFC involved in the most abstract (meta-rule, or time duration) control tasks and operations (Badre & D'Esposito, 2009). Because anterior PFC projects unidirectionally to premotor cortex, as well as bidirectionally with dorsal and ventral lateral prefrontal cortex as well as medial prefrontal cortex (Petrides & Pandya, 2007), it is anatomically reasonable to consider top-down control operations to be involved in both motor and attentional control. This becomes important

later when considering recent advances in neuroimaging of ADHD.

Functional Linkage of Attention and Impulse Control

Both for expository purposes and functionally, the elements of attention and impulse control can be understood as partially distinct. They do not always operate in perfect tandem; they may develop at different rates and they can be measured in distinct ways. In particular, attention can regulate mental processes, even when no motor behavior occurs. While one can speak of cognitive impulsivity (e.g., racing thoughts in hypomania, or inability to suppress thoughts in ADHD), to avoid confusion, regulation of mental contents is treated here under the purview of attention. The term *impulsive* is reserved for overt motor responses and to decision making that results in an associated motor, vocal, or overt emotional response.

Yet this separation of attention and impulse control should not be overstated. Attention and impulse control are closely related both functionally and developmentally. This is important because it is tempting to link problems in regulation of impulse with faulty type 1 processing and problems in regulation of attention with faulty type 2 processing. While doing so can be helpful in some respects, it is a misleading simplification in terms of development or neural function. Attention can regulate impulse and impulse can capture attention. As noted earlier, for a young infant attention is captured involuntarily by an attractive object and she reaches for it automatically (type 1 process). Soon thereafter, a toddler can deliberately look away from a tempting but forbidden object, using attention to regulate impulse as well as emotional arousal (type 2 process). Thus, from early infancy on, attention assists with self-regulation and impulse control (including emotion regulation, behavioral control, and cognitive control) (Pollak & Tolley-Schell, 2004; Rothbart & Rueda, 2005).

Functionally, some kinds of individual variation in impulsivity can be seen as variation in stylistic features of type 2 processing, that is, simply preferences for reflective or nonreflective decision making. Other kinds of individual variation in impulsive decision making are not so easily seen as part of type 2 process, but rather as type 1 process (e.g., intensity of response to incentive). Further, in day-to-day behavior attention and impulse often act in concert or in tension. Both can be influenced, for example, by sensory reflex response. Likewise, both are influenced by strong affective arousal. In the case of strong anxiety (signal of *possible* threat) or fear (*actual* threat), attention is captured and existing impulses are suppressed and replaced by impulses related to the anxiety or fear

(e.g., impulse to flee). In the case of strong excitement due to potential reward, attention is captured and impulses are activated toward the incentive. Likewise, as noted before, the neural systems that support certain attentional skills are closely related to those that support control of motor responses, or impulses (Calkins & Fox, 2002). Further, one can easily think of intersections that do not easily map on to a discrete process model. If the keys are locked in the car because I got distracted as I quickly slammed the door, which impulse control process is faulty? Or is the impulsive action actually an instance of inattention?

Summary Comments About Two-Process Model

An essential qualification is that both attention and impulsivity are multicomponential—neither is a unitary construct, and so neither can be summarily described in terms of the two-process model without further differentiation. The subcomponents of attention, as well as of impulse control, develop dynamically, at different rates, integrating in different ways at different points in development, all reflecting the dynamic nature of neural maturation. In what ensues these component operations and their partially distinct neurobiology are noted. This is done via a taxonomic approach to the various functional domains at hand, as the discussion outlines the different types of regulation and their components relate them to development and to psychopathology.

Therefore, having outlined a fundamental heuristic model, and described the potential to locate developmental breakdowns in a handful of key processes, the essay can now consider in more depth the elements of attention and the elements of impulsivity *per se* and how they might develop or go awry in development.

ATTENTION: CONCEPTUAL FRAMEWORK AND KEY ADHD-RELATED EFFECTS

Definition

By classic definitions, attention refers to the facilitated processing of one source of information over another, although broad usages of the term can include capacity pool (Engle, 2002), as well as response readiness, arousal, regulation (Taylor, 1995), and other abilities (Mirsky & Duncan, 2001). With regard to mechanisms, as outlined by Yantis (1998), attentional selection can be based on spatial location (Eriksen & Hoffman, 1972), object features (Treisman & Gelade, 1980; Wolf, 1998), timing (Shapiro & Raymond, 1994), movement, or other object

characteristics (Kahneman & Henik, 1981). Attentional selection is influenced by two overarching component functions, underscoring a dual-process theme: (1) bottom-up stimulus-driven processes that are relatively automatic and relatively early developing, and (2) top-down goal-driven processes that are strategic, relatively deliberate, related to the concept of executive control, and later developing. Consistent with the overarching framework used here, it has long been recognized that these two types of processes interact continuously, each placing constraints on the other (Schneider & Shiffrin, 1977). Researchers attempt to isolate these two influences in varying ways, depending on part on the age of the child. At the same time, as noted, often overlooked in the purely cognitive approaches to attention is the role of affective motivation in attentional allocation (Nigg & Casey, 2005; Pennington, 2002). Another limitation to this general formulation occurs in very early infancy. While one might see a semblance of exogenous versus endogenous attention in infants, top-down control supported by prefrontal cortex is not possible for very young infants. Their attention is governed by multiple processes, but these do not map neatly onto the two-process model.

However, even without considering emotion or infant development, the complexity of the cognitive component of attention alone has led to a massive literature that cannot be adequately captured in less than book length (de Fockert, 2013; Posner, 2004; Srinivasan, Thorpe, & Nunez, 2013). Highlighted here are certain aspects of attention that have been particularly influential in studies of attentional disorders and ADHD, that may be of most interest to students, or that suggest directions for the future.

Visual attention is the most well-understood modality of attention and therefore most often examined in developmental psychopathology. While not all aspects of visual attention generalize to auditory attention, a useful understanding of the principles of attention can be gained by considering results of visual attention studies and so they are the focus here.

Attention in the Brain

Turning to the brain, one recent perspective suggests six relevant and partially dissociable functional neural networks involved in attention. Figure 14.4 illustrates four of these putative cortical brain networks as recently synthesized from task-based and resting state functional brain imaging literature (Power et al., 2011) and consolidated recently by Petersen and Posner (2012) in an update of their earlier influential model of attention (Posner & Petersen, 1990).

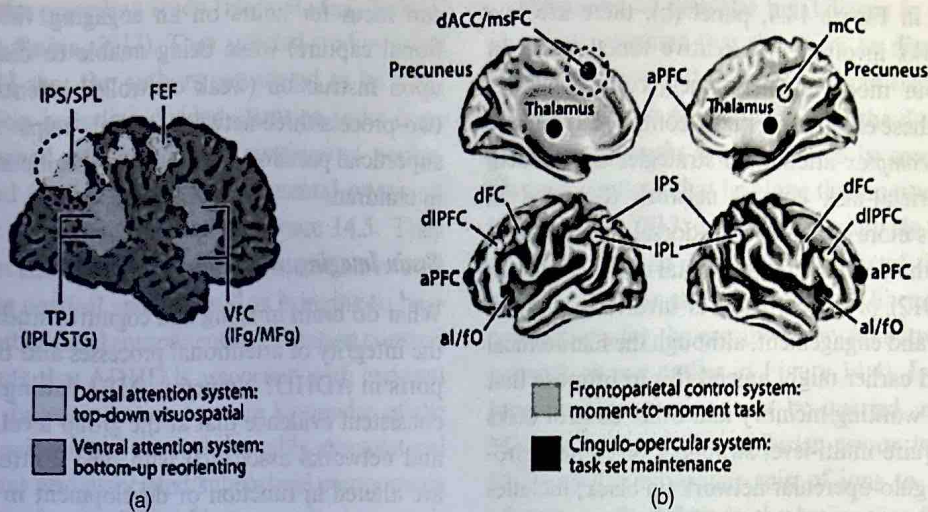


Figure 14.4 Brain attention networks. Panel a: Attention networks. The dorsal attention network = interparietal sulcus (IPS), frontal eye fields (FEF); it is associated with strategic direction of attention. The ventral attention network includes temporal-parietal junction (TPJ), inferior frontal gyrus (IFG). It is associated with bottom-up capture of attention. Panel b: Executive control networks involved in higher order attentional allocation and control on task. Frontoparietal network includes dorsolateral PFC (dlpfc), IPS, medial cingulate (mCC); cingulo-opercular network includes dorsal anterior cingulate (dACC), anterior pole (aPFC), fronto-operculum (AI/FO), and thalamus as well as regions of striatum not shown. These canonical networks are inferred from functional connectivity as well as task based functional MRI data. For detailed discussion see Petersen & Posner (2012). See footnote 1. *Source:* Figure adapted with permission from Petersen, S. E., & Posner, M. I. (2012). The attention system of the human brain: 20 years after. *Annual Review of Neuroscience*, 35, page 76.

Panel (a) depicts two now well-recognized networks for the orienting of visual attention (Corbetta & Shulman, 2002). A dorsal frontoparietal network is engaged during top-down control of attention or goal-directed attention. It includes the frontal eye fields and interparietal sulcus (IPS). A ventral frontoparietal network is engaged during stimulus (or bottom-up) capture of attention and includes the temporoparietal junction (TPJ), the supramarginal gyrus, and inferior frontal gyrus³ (Chica, Bartolomeo, & Lupianez, 2013; Corbetta & Shulman, 2002; Thiebaut de Schotten et al., 2011). Note that while these two networks are well defined, an entire network does not inevitably activate in lock-step during all attentional tasks. For example, in the ventral network, TPJ and inferior frontal gyrus activation can be dissociated based on whether a novel stimulus is expected or unexpected (Shulman et al., 2009). Further, the top-down control of visual versus auditory attention likely share some features of attention (Petersen & Posner, 2012) but also differ somewhat anatomically with regard to perceptual signals. For example, while both involve middle and inferior frontal gyrus, the frontal eye

fields are not as involved in auditory task control (Braga, Wilson, Sharp, Wise, & Leech, 2013). It is crucial to add that additional circuits or loci not shown here are engaged in attention, particularly when attention is motivated. These would include subcortical and basal ganglia brain regions as well as network-specific thalamic nuclei depending on the nature of the visual stimulus. Thus, the focus on cortical networks here is only part of the brain's overall engagement in attention.

The map to type 1 and type 2 processing for these two particular circuits is helpful yet rather incomplete. While most conceptions grant some degree of strategic control of attention to the dorsal network in the sense of endogenous orienting, the orienting systems are primarily involved in allocation of attention, but not more difficult control tasks like suppressing interference, handling response conflict, or complex working memory tasks. For these types of challenges we must invoke additional neural networks.

Experts disagree as to whether executive control is handled by one or two neural networks (Botvinick, Braver, Barch, Carter, & Cohen, 2001; Petersen & Posner, 2012), but these perspectives do not dramatically change the story here. The perspective that argues for two such networks (Dosenbach, Fair, Cohen, Schlaggar, & Petersen, 2008; Petersen & Posner, 2012) proposes the following.

³Some renderings include the frontal operculum (Chica et al., 2013, as cited in text), but here I follow the version provided by Petersen and Posner (2012), who suggested differentiating the ventral network more fully from the fronto-opercular network.

As depicted in Figure 14.4, panel (b), there are two putative networks involved in executive functioning. In keeping with our model of hierarchical control in prefrontal cortex, these executive (or task-control) circuits are engaged when *complex* attentional strategies are needed. The frontal-parietal-task control network (depicted in yellow), which is more anterior to the dorsal attention network, includes the dorsolateral prefrontal cortex. Petersen and Posner (2012) propose that it is involved in initial task evaluation and engagement, although the hierarchical model described earlier might amend this to propose that it is engaged in working memory and other control tasks that do not require multi-level switching or context processing. The cingulo-opercular network (in black) includes the remaining regions usually associated with executive control including the lateral frontal pole, dorsal anterior cingulate, dorsolateral anterior prefrontal cortex, and anterior insula. Petersen and Posner (2012) proposed that it is involved in maintaining effort on a task and suppressing competing information. It may also be engaged in particularly difficult task switching paradigms, according to the hierarchical model. In other words, depending on task complexity, abstraction, and duration, type 2 processes or attention allocation will involve both the dorsal attention and one or both executive control networks.

Two other networks are important that are not shown in Figure 14.4. A so-called *default mode network* is also involved; it includes the posterior cingulate cortex, the ventromedial prefrontal cortex, lateral inferior parietal lobes, and medial temporal structures. Its functions are still debated, but it activates when task control is relaxed and de-activates when task control is engaged. Yet it also is active in many other task situations, so it may be important to some tasks such as determining the relative focus on internal and external stimuli, in retrieving autobiographical information, or in integrating recent experiences (Spreng, 2012). The posterior cingulate cortex in particular may be involved in regulating consistency of responding (Leech & Sharp, 2013) and so may be relevant to resource allocation and response variability, as discussed in subsequent sections. Notably, the default mode network appears to have altered connectivity in ADHD as well as other disorders (Fair et al., 2012). The final network is an alerting (or, in some conceptions, arousal) network, which is discussed subsequently under the section heading, *capacity*.

This multi-network model helps us heuristically think about the ADHD clinical picture. For example, a common clinical observation in ADHD is that the child cannot focus on their homework or on what they are being told to do (weak controlled attention), but they

can focus for hours on an engaging videogame (attentional capture) while being unable to disengage from it upon instruction (weak controlled attention). Thus, the two-process/three-network model helps us resolve the superficial paradox of situational-specific attentional focus in children.

Brain Imaging, Attention, and ADHD

What do brain imaging and cognitive studies tell us about the integrity of attentional processes and their neural supports in ADHD? Numerous MRI findings have provided consistent evidence that at the group level, neural regions and networks associated with type 2 attention processes are altered in function or development in groups of children with ADHD (Bush, 2010; Durston, van Belle, & de Zeeuw, 2011; Rubia, 2011). Key findings from MRI studies of the brain in recent years are noted here, with an emphasis on meta-analytic findings when available. Then, studies of attention itself (not of brain function), conducted in the laboratory under controlled conditions, are considered in relation to ADHD.

Functional MRI studies are quite interesting in regard to the two-process model and the circuits just described. A meta-analysis (Cortese et al., 2012) covering data through 2011 identified 55 relevant functional MRI studies with a total of 741 ADHD and 801 comparison subjects. That meta-analysis unfortunately was quite limited in that it did not report effect sizes. However, it did conclude that ADHD was associated with abnormal (mostly hypoactivation) response in multiple brain nodes that are part of distinct brain circuits. While it relied on canonical networks suggested from a study of 1,000 adults (Yeo et al., 2011), these are similar to those described in Figure 14.4. Cortese et al. concluded that in addition to frontal-striatal circuitry, in children there was abnormal hypoactivation in the ventral attention network, while in adults there was abnormal hypoactivation in the dorsal attention network. These results were not explained by either comorbid conditions or treatment history. In addition, they highlighted that findings in a smaller number of studies related to hyperactivation of the default network as well as sensory networks (vision) and hypoactivation of motor networks that need further exploration. Again, it is important to point out that the method used in this analysis relied on so-called functional networks that are being explored in the brain using MRI; these are confined to cortical networks and do not represent thalamic, limbic, or other subcortical components of these and other networks in the brain.

A second meta-analysis took a different approach, enabling consideration of subcortical areas and thus filling

in a gap from the preceding study (Hart, Radua, Nakao, Mataix-Cols, & Rubia, 2013). They selected studies using task-based MRI that the authors considered to be measures of attention including divided attention tasks (e.g., oddball), sustained attention (e.g., continuous performance test), and flexible attention (e.g., mental rotation). The results are reproduced in part in Figure 14.5. They further highlight the role of breakdown in attention-related neural circuits in cortical areas, as well as bringing to bear the important subcortical components of attention control. The figure depicts that ADHD is associated with regional hypoactivation during attention tasks in key nodes of the frontoparietal task control network (notably, dorsolateral prefrontal cortex) and associated subcortical components of thalamo-cortical-striatal neural loops involved in cognitive control, including putamen, caudate, and portions of the thalamus. It also shows altered activation in the insula and inferior parietal lobe, in the precuneus, and increased activation in the left cuneus and right cerebellum.

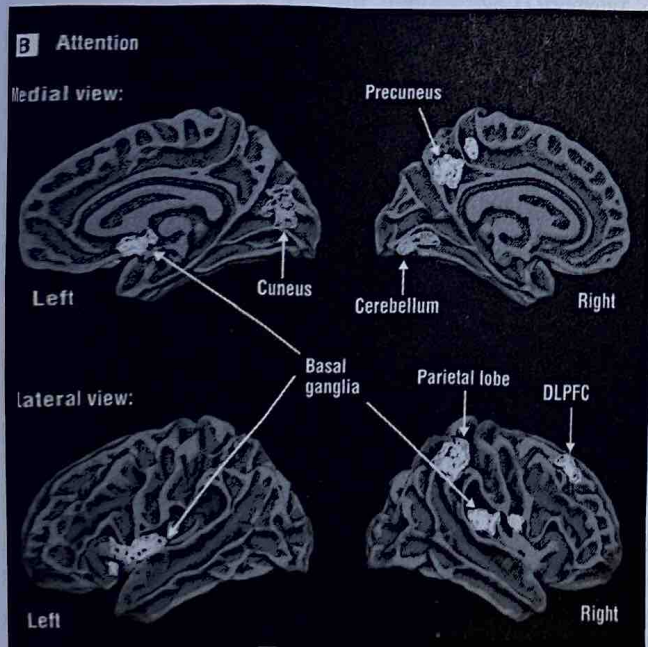


Figure 14.5 Altered brain activation during attention tasks in ADHD. ADHD associated on attention tasks with decreased activation in the right dorsolateral prefrontal cortex (DLPFC), left putamen and globus pallidus, right posterior thalamus (pulvinar) and caudate tail extending into the posterior insula, in the right inferior parietal lobe, and in the precuneus and superior temporal lobe. Increased activation in the left cuneus and in the right cerebellum. See footnote 1. *Source:* Adapted with permission from H. Hart, J. Radua, T. Nakao, D. Mataix-Cols, & K. Rubia (2013), Meta-analysis of functional magnetic resonance imaging studies of inhibition and attention in attention-deficit/hyperactivity disorder: Exploring task-specific, stimulant medication, and age effects, *Journal of the American Medical Association Psychiatry*, 70(2), page 191.

Whenever we consider breakdowns in neural systems, we must recognize that they may be disrupted via their interconnecting pathways rather than their modular loci. For example the distributed loci of the dorsal and ventral attention networks are thought to be connected via long fiber connections that lie along the longitudinal fasciculus (Chica et al., 2013) and that speculatively may participate in connecting the distributed regions of these attentional networks. However, note that these fibers may also form connections for the executive control network (the fourth network shown earlier in Figure 14.4). Integrity of these long fiber connections can be queried using a different MRI technique called *diffusion tensor imaging* or DTI. DTI uses the molecular spin of ions to gauge direction of water or fluid flow in the brain; that flow tends to be directional where axons are well developed and diffuse where they are less developed. Leaving aside here several complexities in interpreting that signal, from a simplified point of view we can consider the degree of directional water flow as an index of axonal integrity at the level of white matter microstructure. Several studies have taken this approach with ADHD.

A meta-analysis is again helpful, pooling multiple studies of diffusion tensor signal (measuring white matter development) in ADHD children and adults. Although the collective database remains small (9 studies, 173 ADHD and 169 control participants from 5 to 49 years of age), one consistent finding in research on ADHD in the past decade has been alterations in white matter microstructure along the superior longitudinal fasciculus (van Ewijk, Heslenfeld, Zwiers, Buitelaar, & Oosterlaan, 2012). Figure 14.6 reproduces their key results. Findings that held across the pooled data are highlighted with blue coloring available online. Note the second column in the figure, showing effects for the right anterior corona radiata. This was the largest cluster found, and as the authors noted probably includes parts of the superior longitudinal fasciculus, implicating altered development of fibers that support top-down control of attention via either the dorsal attention network or the frontal-parietal task control network. It was notable that there were stronger effects in younger children (Nagel et al., 2011), suggesting the potential to use this technique to map development of ADHD.

Laboratory Tests, Attention, and ADHD: Type 1 Attention

Laboratory tests of attention are illuminating but the literature is vast. Type 1 attentional regulation can be studied via experiments to assess reflexive orienting and perceptual selection, among others. Other experiments have tapped